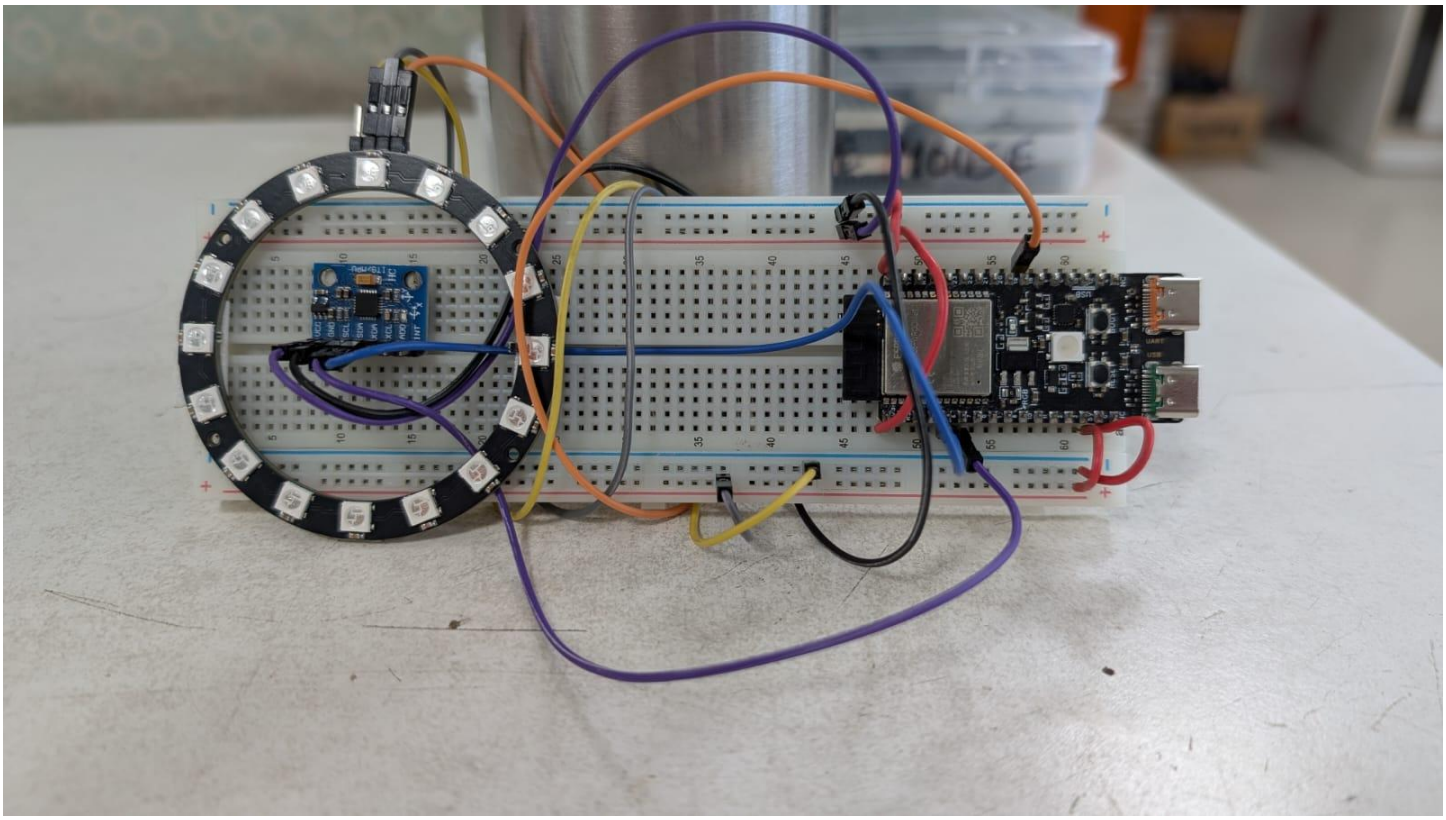
Rolling Marble Simulation Playbook

I. Challenge

Design and execute an embedded Hardware-in-the-Loop (HIL) physics engine that models the dynamic kinematics of a virtual sphere rolling along a circular path. The system must translate continuous spatial data from an inertial sensor into dynamic physical properties—such as inertia, momentum, friction, and mass—and map the resulting high-precision floating-point coordinates onto a discrete 16-LED addressable display ring with zero latency at a steady 50 Hz execution loop.



II. Guide

A. Concept

To implement a responsive, realistic physics simulation on a microcontroller, we apply these key trigonometric and classical mechanics concepts:

- **1. Downhill Incline Resolver:** Micro-electromechanical system (MEMS) accelerometers measure raw gravitational vectors along spatial axes. To isolate the true direction that gravity pulls a physical object downwards, the system relies on the two-argument arctangent function:

$$\theta_{\text{target}} = \text{atan2}\left(x_{\text{Angle}}, y_{\text{Angle}}\right) \times \frac{180}{\pi}$$

This evaluates the quadrant signs to map the absolute tilt target from $0°$ to $360°$.

  - *Learn more:* The Mathematics of the Atan2 Function

- **Modular Angular Distance & Wrap-Around:** Direct subtraction ($\theta_{\text{target}} - \theta_{\text{marble}}$) breaks down across the $0°/360°$ boundary, causing false rotational directions. The physics engine constrains angular error to $[-180°, 180°]$ to ensure the simulated marble always takes the shortest physical path around the circular track.
    - *Learn More:* Stack Overflow – Finding the Shortest Distance Between Two Angles
- **3. Semi-Implicit Euler Numerical Integration:** Instead of complex continuous equations, the physics engine updates the particle across tiny discrete slices of time ($\Delta t = 20$ ms). Restoring torque yields acceleration, which modifies the current velocity before being compressed by a damping factor (μ) representing surface friction:

$$v_t = (v_{t-1} + a) \times \mu$$

$$x_t = x_{t-1} + v_t$$

    - *Learn more:* Numerical Integration
- **4. Continuous-to-Discrete Remapping:** The physics core operates entirely with high-precision floating-point variables to maintain momentum accuracy. To display this on hardware, the continuous angle must be scaled, rounded to the nearest integer index, and wrapped using a modulo operation to light the correct pixel:

$$\text{Index}_{\text{LED}} = \left\lfloor \text{round}\left(\frac{\theta_{\text{marble}}}{360/N_{\text{LEDs}}}\right) \right\rfloor \pmod{N_{\text{LEDs}}}$$

    - *Learn more:* Quantization Error Explained

Real-World Example: Advanced stability control systems in aerospace rocketry, smartphone screen orientation adjustments, and active anti-sway suspension structures in high-performance automotive platforms. They sample real-time tilt vectors to shift virtual or mechanical elements against gravity.

B. Components

- **Microcontroller:** WeAct Studio ESP32-C6 Dev Board.
- **Motion Sensor:** MPU6050 6-DoF Inertial Measurement Unit.
- **Visualizer Array:** 16-element WS2812B NeoPixel RGB LED Ring.
- **Interconnect Base:** Solderless breadboard and jumper hookup wires. Double-sided tape is purely optional for cosmetic reasons to place MPU6050 and RGB LED Ring module on the breadboard.

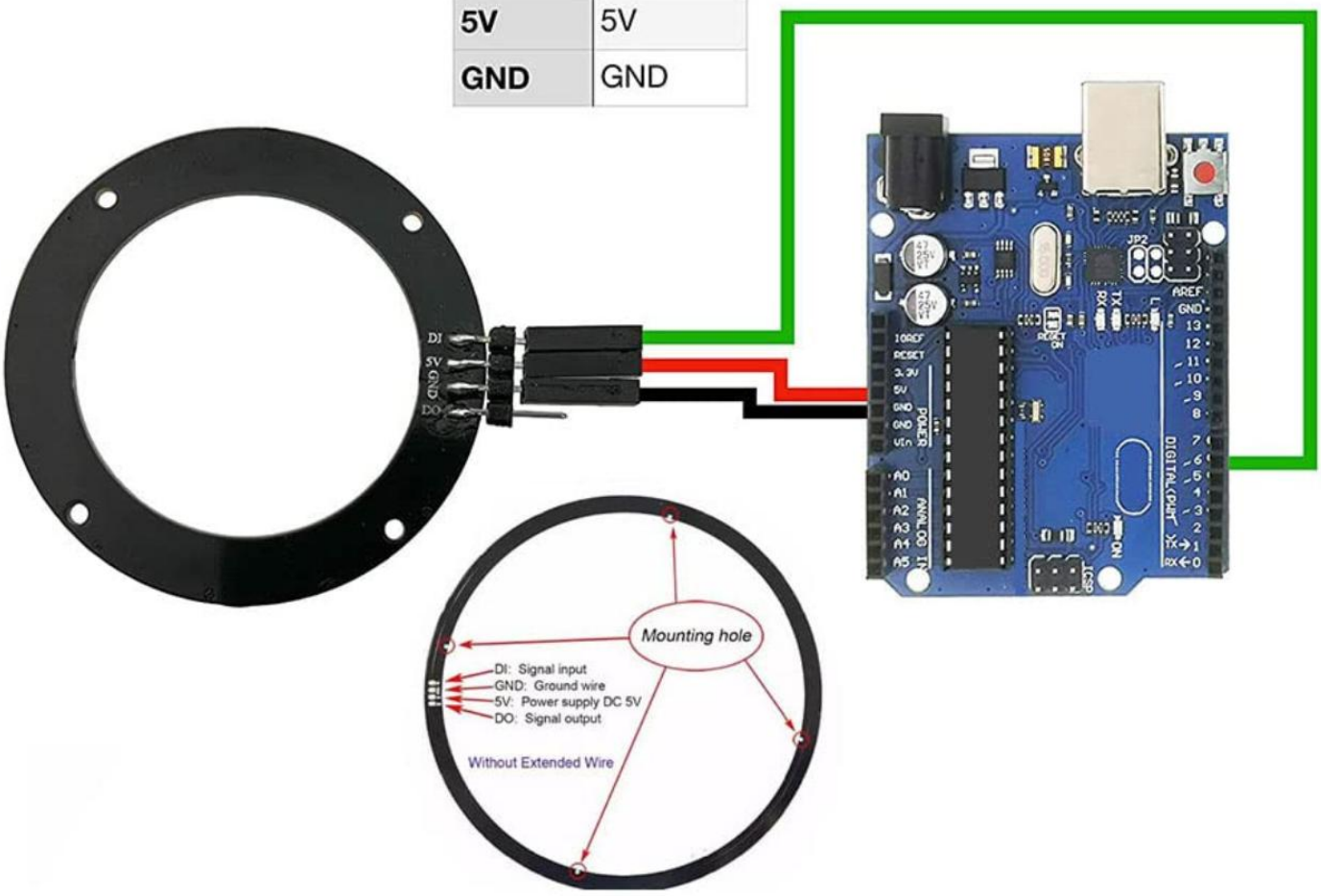
C. Connections

The system uses an I^2C protocol channel to extract the sensor readings and a high-speed one-wire pulse control channel to update the lighting ring:

Source Device	Pin Label	Target MCU Pin (ESP32-C6)	Operational Functional Description
MPU6050 IMU	VCC	3.3V	Logic Power Supply
	GND	Gnd	Common Ground Reference
	SDA	GPIO 6	Hardware I^2C Serial Data Communication
	SCL	GPIO 7	Hardware I^2C Serial Clock Line
NeoPixel Ring	5V	V5 / VIN	High-Current Input (Direct USB 5V Rail)
	GND	Gnd	Common Ground Reference
	DI	GPIO 18	High-Speed Digital Pixel Control Input
	DO	Unconnected	Open Daisy-Chain Output Pass-through

How to connect RGB LED Ring Module is explained as shown by this diagram:

Board	LED RING
Pin6	DI
5V	5V
GND	GND



D. Code

This program sets up the hardware lines, calibrates the gyroscope base offset, and runs the integration equations at a consistent 50 Hz step interval:

```
#include <Wire.h>
#include <MPU6050_tockn.h>
#include <Adafruit_NeoPixel.h>

#define LED_PIN     18
#define NUM_LEDS    16
#define I2C_SDA     6
#define I2C_SCL     7

MPU6050 mpu6050(Wire);
Adafruit_NeoPixel ring(NUM_LEDS, LED_PIN, NEO_GRB + NEO_KHZ800);

// --- PHYSICS ENGINE VARIABLES ---
float marblePos = 0.0;       // Current position of the marble in degrees (0.0 to 360.0)
float marbleVelocity = 0.0;  // Current angular velocity
```

```
const float FRICTION = 0.92; // Damping factor (0.99 = ice/no friction, 0.80 = heavy mud)
const float GRAVITY_MULT = 0.02; // How responsive the marble is to gravity/weight

void setup() {
  Serial.begin(115200);
  Wire.begin(I2C_SDA, I2C_SCL);

  ring.begin();
  ring.setBrightness(40);
  ring.show();

  Serial.println("Place flat for IMU calibration...");
  mpu6050.begin();
  mpu6050.calcGyroOffsets(true);
  Serial.println("Physics Engine Ready!");
}

void loop() {
  mpu6050.update();
  float xAngle = mpu6050.getAngleX();
  float yAngle = mpu6050.getAngleY();

  // 1. Calculate overall tilt steepness (Magnitude)
  float tiltMagnitude = sqrt((xAngle * xAngle) + (yAngle * yAngle));

  // 2. Only update physics if the board is actually tilted
  if (tiltMagnitude > 2.0) {
    // Find the downhill target angle where gravity pulls a marble (opposite of bubble level)
    float targetAngle = atan2(xAngle, yAngle) * 180.0 / M_PI;
    if (targetAngle < 0) targetAngle += 360.0;

    // 3. Calculate shortest angular distance to target (handles 0/360 wrap-around)
    float angularError = targetAngle - marblePos;
    if (angularError > 180.0)  angularError -= 360.0;
    if (angularError < -180.0) angularError += 360.0;

    // 4. Euler Integration: Accel -> Velocity -> Position
    float acceleration = angularError * tiltMagnitude * GRAVITY_MULT;

    marbleVelocity += acceleration;
    marbleVelocity *= FRICTION; // Apply simulated surface resistance
    marblePos += marbleVelocity;

    // Keep marble position strictly within 0-360 degrees
    if (marblePos >= 360.0) marblePos -= 360.0;
    if (marblePos < 0.0)    marblePos += 360.0;
  } else {
    // If the board is perfectly flat, slowly bleed off any remaining velocity
    marbleVelocity *= 0.5;
    marblePos += marbleVelocity;
  }

  // 5. Map the float position angle to the nearest integer LED index
  int currentLED = ((int)round(marblePos / (360.0 / NUM_LEDS))) % NUM_LEDS;

  // 6. Render the Marble
  ring.clear();

  // Draw the main marble (Bright Blue)
  ring.setPixelColor(currentLED, ring.Color(0, 150, 255));

  // Draw a dim "tail" or motion blur behind the marble based on velocity direction
  int tailLED = (marbleVelocity > 0) ? (currentLED - 1 + NUM_LEDS) % NUM_LEDS : (currentLED + 1) % NUM_LEDS;
  if (abs(marbleVelocity) > 2.0) {
    ring.setPixelColor(tailLED, ring.Color(0, 40, 100));
  }
  ring.show();

  delay(20); // Steady 50Hz frame rate for smooth physics integration
}

```

III. Play & Share

- **Introduce Elastic Boundary Collisions:** Instead of allowing the marble to spin continuously all the way around the ring, create a virtual brick barrier at LED index 0 and 8. Modify your integration code so that if the marble's position hits a barrier, the position stops instantly, and the velocity multiplies by -0.6 to bounce backwards!
- **Simulate Multi-Body Physics:** Expand your array parameters to introduce a second marble onto the track. Can you write a software check that senses when both indexes share the same discrete space and performs an **elastic momentum swap** so the particles bump away from each other?
- **Design a Dynamic Maze Game:** Use the code to build a game where a player tilts the board to guide the marble into a designated, blinking "target gate" LED within a specific time limit while avoiding dynamic, red "trap" LEDs that reset the score.